CO1 ASSESSMENT TOOL-1

course code	CSA0702	course Tittle	computer Networks
CO Assessed	CO1	Assessment	Tool-short Answers
weightage	40 %	Total marks	20 marks

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Total marks	20	obtained marks	19

understanding of concept	explanation	Application	organization	Accuracy	clarity
25 %	25 %	20 %	15 %	10 %	5 %
5	5	4	3	1	1

ourse : Computer Networks
code : CSA0702
Slot : B

CO1 - ASSESSMENT-1

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1) File Transport or Transfer using Transport and Data Link layers

Given :

- File size = 150 MB = ~~bytes~~ 150×10^6 bytes
- segment size = 1500 bytes
- Frame overhead = 40 bytes
- Bandwidth = 50 Mbps

step 1 : Number of segments

$$\text{segments} = \frac{150 \times 10^6}{1500} = 100000$$

step 2 : Number of frames

Each segment = 1 frame

Frames = 100000

step 3 : Total overhead

overhead = 100000 × 40
= 4,000,000 bytes
= 4 MB

step 4 : Total Data Transmitted

Total Data = 150 + 4 = 154 MB

convert to bits :

154 × 8 = 1232 Mb

step 5 : Transmission Time

$$\text{Time} = \frac{1232}{50} = 24.64 \text{ seconds}$$

Final Answer :

- segments = 100,000
- Frames = 100,000
- overhead = 4MB
- Transmission Time = 24.64 sec

Explanation

- Transport Layer : divides data into segments and ensures reliability using error control and flow control.

- Data Link Layer : Adds headers/trailers, detects errors, and ensures frame delivery.

2) sliding window protocol

Given :

- window size = 6 frames
- Total frames = 48
- Each window loses 1 frame

step 1 : Number of windows

$$\text{windows} = \frac{48}{6} = 8$$

step 2 : Total Retransmissions

Each window → 1 frame lost

Retransmissions = 8 × 1 = 8

Final Answer:

Number of windows = 8

Total retransmissions = 8 frames

Explanation:

- Flow control: ensures sender does not send data faster than receiver can handle.

- sliding window improves efficiency by sending multiple frames before waiting for acknowledgment.

- Retransmission handles lost frames, ensuring reliable communication.

3) IP Fragmentation

Given:

- ~~Packet size = 12,000~~ bytes
- MTU = 1500 bytes
- Header per fragment = 20 bytes

step 1: Data per Fragment

Data per fragmement = 1500 - 20
= 1480 bytes.

step 2: Apply 8-byte Rule

In IP fragmentation, data must be a multiple of 8 bytes.

$\frac{1480}{8} = 185$ (valid)

So, 1480 bytes is a acceptable.

step 3 : Number of fragments

$$\text{Fragments} = \frac{12000}{1480} = 8.1 \Rightarrow 9 \text{ fragments}$$

step 4 : Total Header overhead

$$\text{overhead} = 9 \times 20 = 180 \text{ bytes}$$

Final Answer :

- Number of fragments = 9
- Total header overhead = 180 bytes

Explanation :

- Networ Layer performs fragmentation when packet size exceeds MTU.
- Each fragment carries its own header.
- Destination system reassembles fragments using identificat and offset fields.
- The 8-byte alignment rule ensures proper reassembly.

4) sliding window protocol

Given :

- window size = 12 frames
- Total frames = 48
- one frame lost per window

step 1 : Number of windows

$$\text{windows} = \frac{48}{12} = 4$$

step 2 : Total Retransmissions

Each window → 1 frame lost

Retransmissions $= 4 \times 1 = 4$

Final Answer :

- Number of windows $= 4$
- Total retransmissions $=$ 4 frames.

Explanation :

- Larger window size → fewer windows → better efficiency.
- Flow control ensures smooth data transfer without congestion.
- Lost frames are retransmitted → ensures relaible communication.

5) session layer synchronization

Given :

- Total file size $= 600$ MB
- checkpoint After every $= 60$ MB
- Failure occurs at $= 390$ MB

step 1 : Number of checkpoints created

checkpoints $= \frac{390}{60} = 6.5$

only completed check points are counted → 6 checkpoints

(At : 60, 120, 180, 240, 300, 360 MB)

step2 : Data to be Retransmitted

Last checkpoint = 360 MB

Retransmission Data = 390 − 360 = 30 MB

Final Answer :

- Number of checkpoints = 6
- Data to retransmit = 30 MB

Explanation :

- session Layer uses checkpoints to avoid full retransmission
- After failure, transmission resumes from last checkpoint
- This improves efficiency and reliability in data transfer

6) presentation layer compression and encryption

Given :

- original file size = 900 MB
- compression = 40 %
- Encryption overhead = 5 %

step 1 : compressed file size

$$\text{compressed size} = 900 \times (1 - 0.40)$$
$$= 900 \times 0.65$$
$$= 540 \text{ MB}$$

step2 : Encrypted file size

$$\text{encrypted size} = 540 \times (1 + 0.05) = 567 \text{ MB}$$

step 3 : Tot
Redu

Step 3: Total Percentage Reduction

$$\text{Reduction} = \frac{900-567}{900} \times 100 = \frac{333}{900} \times 100 \approx 37\ \%$$

Final Answer:

- compressed size = 540 MB
- Encrypted size = 567 MB
- Total reduction ≈ 37 %

Explanation:

- presentation layer compresses data → reduces size → faster transmission.
- It also encrypts data → improves security.
- overall, it ensures efficient and secure communication.

7) FTP Application Data Transfer

Given:

- Total data = 3 GB = 3000 MB
- Each request = 3 MB
- Throughput = 120 Mbps

step 1: Number of Requests

$$\text{Requests} = \frac{3000}{3} = 1000$$

step 2: convert Data to Bits

$$3000\,MB \times 8 = 24000\,Mb$$

step 3 : Transmission Time

$$\text{Time} = \frac{24000}{120} = 200 \text{ seconds}$$

Final Answer :

- Number of requests = 1000
- Transmission time = 200 seconds

Explanation :

- Application Layer divides data into smaller requests (FTP chunks)
- It ensures reliable file tranfer, user interaction, and error handling.
- Provides efficient and user-friendly communication services

8) End-to-End Delay

Given :

- Application Layer delay = 4ms
- Transport Layer delay = 3ms
- Network Layer delay = 5ms
- Data Link layer delay = 2ms
- physical Layer delay = 1ms
- Propagation delay = 18ms
- Transmission delay = 12ms

step 1: Total Processing Delay

$$4+3+5+2+1 = 15\text{ms}$$

…2: Total End-to-End Delay

Total Delay = $15 + 18 + 12 = 45$ ms

…al Answer:

Total end-to-end delay = 45 ms

…planation:
All layers together ensure smooth and reliable communication.

Protocol overhead and efficiency
…iven:

useful data = 80 MB = 80×10^6 bytes

Frame size = 1600 bytes

overhead performance = 80 bytes

step 1: useful data per frame

useful data per frame = $1600 - 80 = 1520$ bytes

step 2: Number of frames

$$\text{Frames} = \frac{80 \times 10^6}{1520} \approx 52631.6 \Rightarrow 52632 \text{ frames}$$

step 3: Total overhead

Total overhead = $52632 \times 8 \approx 4.21$ MB

step 4: Transmission Efficiency

$$\text{Efficiency} = \frac{\text{useful Data}}{\text{Total Frame size}} \times 100 = 95\%$$

Final Answer:

- Number of frames = 52,632
- Total overhead ≈ 4.21 MB
- Transmission efficiency = 95%

explanation:

- Each frame carries both data + overhead.
- More overhead → reduces efficiency.
- Efficient protocols aim to maximize useful data transmission while maintaining reliability.

10) Healthcare Network Data transfer

Given:

- original data = 240 MB
- compression = 30 %.
- segment size = 1400 bytes
- overhead perframe = 60 bytes

compressed File size

compressed size = $240 \times (1-0.30) = 168$ MB

Convert to Bytes

168 MB $= 168 \times 10^6$ bytes

Number of segments

$$\text{segments} = \frac{168 \times 10^6}{1400} = 120000$$

Total overhead

overhead $= 120000 \times 60 = 7.2$ MB

Final Data Transmitted

Total Data $= 168 + 7.2 = 175.2$ MB

Final Answer

compressed size = 168 MB

Number of segments = 120,000

Total overhead = 7.2 MB

Final transmitted data = 175.2 MB